2 A particle in a medium is oscillating because of the passage of a transverse wave W_1.

The wave has intensity I at this point. The amplitude of the oscillation is A.

Fig. 2.1 shows the variation with time t of the displacement x of the particle.

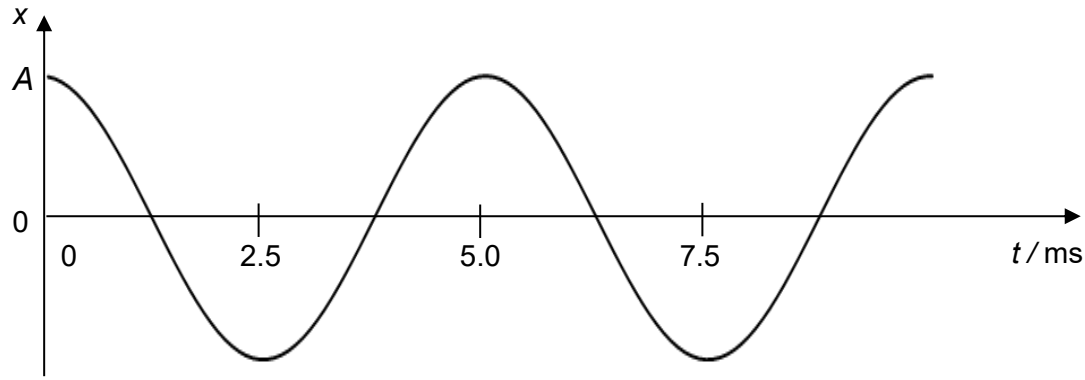

Fig. 2.1

A second, similar transverse wave W_2 has the same frequency and is incident on the same particle. The amplitude of the oscillation due to W_2 alone is $\frac{5}{2}A$ at this point.

(a) Calculate

(i) the frequency of the waves,

frequency = ... Hz [1]

(ii) the intensity, in terms of I, of the wave W_2.

intensity = ...I [2]

(b) **(i)** State two conditions which are necessary for the waves W_1 and W_2 to produce an observable interference pattern.

.....

..... [2]

(ii) State the condition that must be satisfied if the waves are to interfere to produce a minimum resultant intensity at a point.

.....

..... [1]

(iii) Calculate, in terms of I, this minimum intensity.

minimum intensity = ...I [2]